

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 6. Approximation and fitting

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Norm approximation

$$\text{minimize } \|Ax - b\|$$

($A \in \mathbf{R}^{m \times n}$ with $m \geq n$, $\|\cdot\|$ is a norm on $\mathbf{R}^m$)

interpretations of solution $x^* = \operatorname{argmin}_x \|Ax - b\|$:

- **geometric:** Ax^* is point in $\mathcal{R}(A)$ closest to b
- **estimation:** linear measurement model

$$y = Ax + v$$

y are measurements, x is unknown, v is measurement error

given $y = b$, best guess of x is x^*

- **optimal design:** x are design variables (input), Ax is result (output)
 x^* is design that best approximates desired result b

examples

- least-squares approximation ($\|\cdot\|_2$): solution satisfies normal equations

$$A^T A x = A^T b$$

$$(x^* = (A^T A)^{-1} A^T b \text{ if } \mathbf{rank} A = n)$$

- Chebyshev approximation ($\|\cdot\|_\infty$): can be solved as an LP

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & -t\mathbf{1} \preceq Ax - b \preceq t\mathbf{1} \end{array}$$

- sum of absolute residuals approximation ($\|\cdot\|_1$): can be solved as an LP

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T y \\ \text{subject to} & -y \preceq Ax - b \preceq y \end{array}$$

Penalty function approximation

ℓ_p norm, $1 \leq p < \infty$

$$\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p} \Rightarrow \|x\|_p^p = (|x_1|^p + \dots + |x_n|^p)$$

- Replace the object function;
- The new objective function is separable and symmetric;
- We can further generalize this to a penalty function form.

Penalty function approximation

$$\begin{array}{ll} \text{minimize} & \phi(r_1) + \cdots + \phi(r_m) \\ \text{subject to} & r = Ax - b \end{array}$$

($A \in \mathbf{R}^{m \times n}$, $\phi : \mathbf{R} \rightarrow \mathbf{R}$ is a convex penalty function)

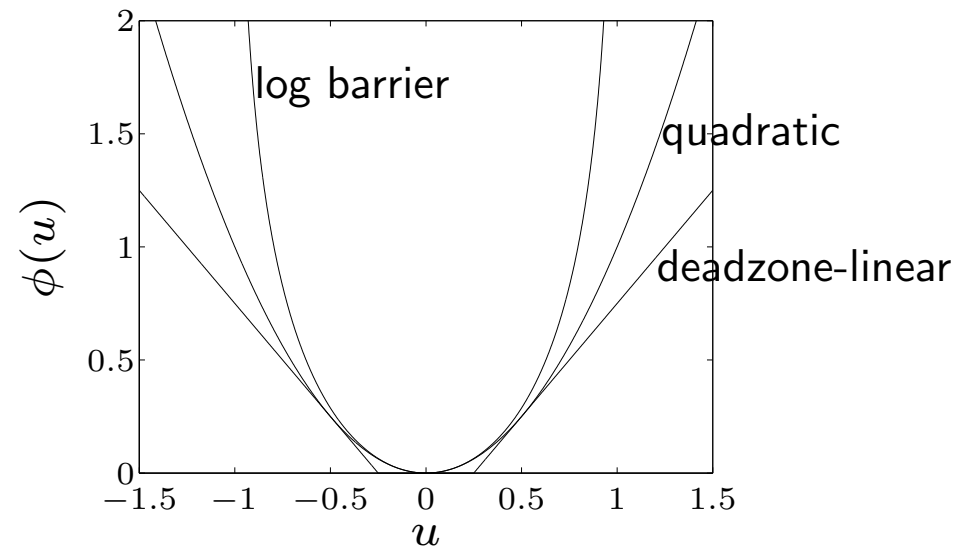
examples

- quadratic: $\phi(u) = u^2$
- deadzone-linear with width a :

$$\phi(u) = \max\{0, |u| - a\}$$

- log-barrier with limit a :

$$\phi(u) = \begin{cases} -a^2 \log(1 - (u/a)^2) & |u| < a \\ \infty & \text{otherwise} \end{cases}$$



Penalty function approximation

ℓ_p norm, $1 \leq p < \infty$

$$\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p} \Rightarrow \|x\|_p^p = (|x_1|^p + \dots + |x_n|^p)$$

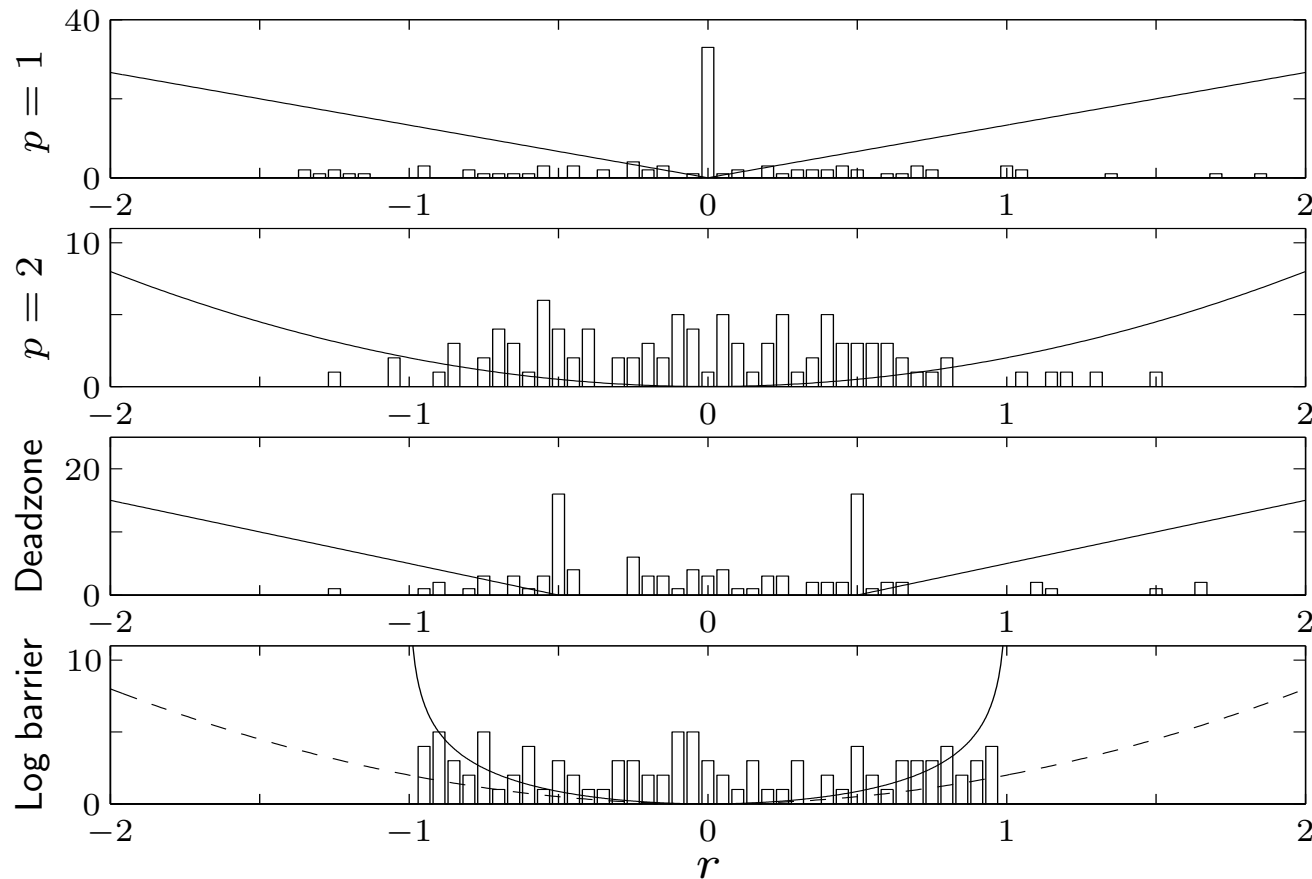
- Replace the object function;
- The new objective function is separable and symmetric;
- We can further generalize this to a penalty function form.

Interpretation of the penalty function:

- The cost $\phi(u_i)$ reflects our dislike of the residue value u_i ;
- Scaling the cost functions are irrelevant;
- Different functions assign different preferences for large or small residues.

example ($m = 100$, $n = 30$): histogram of residuals for penalties

$$\phi(u) = |u|, \quad \phi(u) = u^2, \quad \phi(u) = \max\{0, |u| - a\}, \quad \phi(u) = -\log(1 - u^2)$$



shape of penalty function has large effect on distribution of residuals

Outliers

Outliers cause large residue errors, and we wish to ignore them

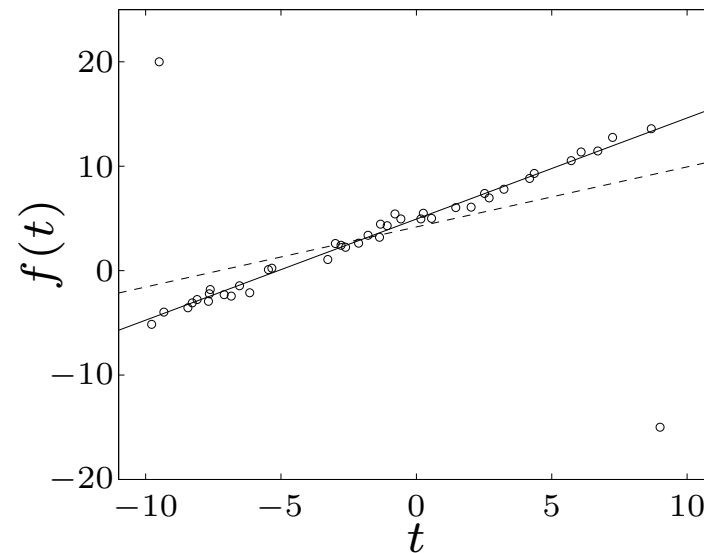
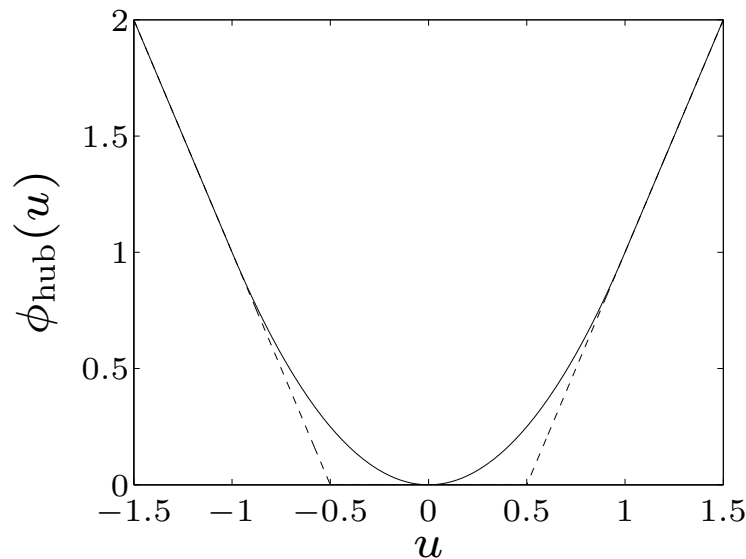
$$\phi(u) = \begin{cases} u^2 & |u| \leq M \\ M^2 & |u| > M. \end{cases}$$

But this function is not convex: use a convex function least sensitive to outliers

Huber penalty function (with parameter M)

$$\phi_{\text{hub}}(u) = \begin{cases} u^2 & |u| \leq M \\ M(2|u| - M) & |u| > M \end{cases}$$

linear growth for large u makes approximation less sensitive to outliers



- left: Huber penalty for $M = 1$
- right: affine function $f(t) = \alpha + \beta t$ fitted to 42 points t_i, y_i (circles) using quadratic (dashed) and Huber (solid) penalty

Sparsity

The sparsest solution:

- A solution $Ax = b$ that has the smallest number of non-zeros;

Sparsity

The sparsest solution:

- A solution $Ax = b$ that has the smallest number of non-zeros;
- Sometimes called the ℓ_0 norm, although it is not a norm.
- ℓ_1 norm is sometimes a good approximation of the ℓ_0 norm (compressive sensing).

Exercise

Formulate the approximation problem to an efficient form

- Minimize $\sum_{i=1}^m \phi(a_i^T x - b_i)$, where $\phi(\cdot)$ is the deadzone-linear penalty function.

Least-norm problems

$$\begin{array}{ll} \text{minimize} & \|x\| \\ \text{subject to} & Ax = b \end{array}$$

($A \in \mathbf{R}^{m \times n}$ with $m \leq n$, $\|\cdot\|$ is a norm on $\mathbf{R}^n$)

interpretations of solution $x^* = \operatorname{argmin}_{Ax=b} \|x\|$:

- **geometric:** x^* is point in affine set $\{x \mid Ax = b\}$ with minimum distance to 0
- **estimation:** $b = Ax$ are (perfect) measurements of x ; x^* is smallest ('most plausible') estimate consistent with measurements
- **design:** x are design variables (inputs); b are required results (outputs)
 x^* is smallest ('most efficient') design that satisfies requirements

examples

- least-squares solution of linear equations ($\|\cdot\|_2$):
can be solved via optimality conditions

$$2x + A^T \nu = 0, \quad Ax = b$$

- minimum sum of absolute values ($\|\cdot\|_1$): can be solved as an LP

$$\begin{array}{ll} \text{minimize} & \mathbf{1}^T y \\ \text{subject to} & -y \preceq x \preceq y, \quad Ax = b \end{array}$$

tends to produce sparse solution $x^\star$

extension: least-penalty problem

$$\begin{array}{ll} \text{minimize} & \phi(x_1) + \cdots + \phi(x_n) \\ \text{subject to} & Ax = b \end{array}$$

$\phi : \mathbf{R} \rightarrow \mathbf{R}$ is convex penalty function

Regularized approximation

$$\text{minimize (w.r.t. } \mathbf{R}_+^2 \text{)} \quad (\|Ax - b\|, \|x\|)$$

$A \in \mathbf{R}^{m \times n}$, norms on $\mathbf{R}^m$ and $\mathbf{R}^n$ can be different

interpretation: find good approximation $Ax \approx b$ with small x

- **estimation:** linear measurement model $y = Ax + v$, with prior knowledge that $\|x\|$ is small
- **optimal design:** small x is cheaper or more efficient, or the linear model $y = Ax$ is only valid for small x
- **robust approximation:** good approximation $Ax \approx b$ with small x is less sensitive to errors in A than good approximation with large x

Vector optimization

general vector optimization problem

$$\begin{array}{ll} \text{minimize (w.r.t. } K) & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p \end{array}$$

vector objective $f_0 : \mathbf{R}^n \rightarrow \mathbf{R}^q$, minimized w.r.t. proper cone $K \in \mathbf{R}^q$

convex vector optimization problem

$$\begin{array}{ll} \text{minimize (w.r.t. } K) & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

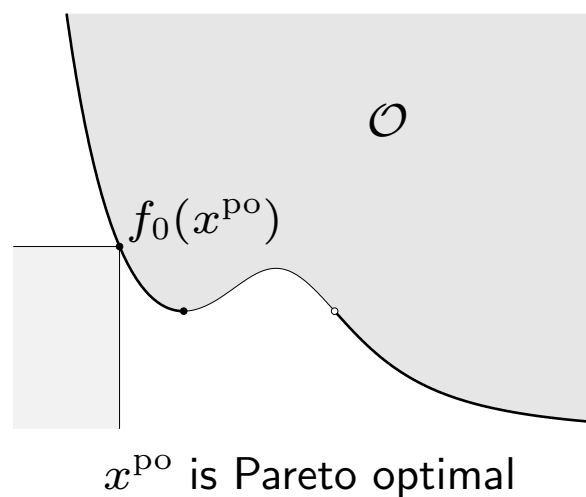
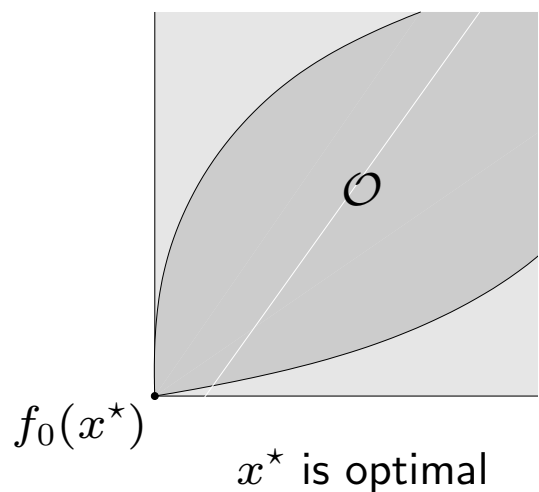
with f_0 K -convex, $f_1, \dots, f_m$ convex

Optimal and Pareto optimal points

set of achievable objective values

$$\mathcal{O} = \{f_0(x) \mid x \text{ feasible}\}$$

- feasible x is **optimal** if $f_0(x)$ is the minimum value of $\mathcal{O}$
- feasible x is **Pareto optimal** if $f_0(x)$ is a minimal value of $\mathcal{O}$



Multicriterion optimization

vector optimization problem with $K = \mathbf{R}_+^q$

$$f_0(x) = (F_1(x), \dots, F_q(x))$$

- q different objectives F_i ; roughly speaking we want all F_i 's to be small
- feasible x^* is optimal if

$$y \text{ feasible} \implies f_0(x^*) \preceq f_0(y)$$

if there exists an optimal point, the objectives are noncompeting

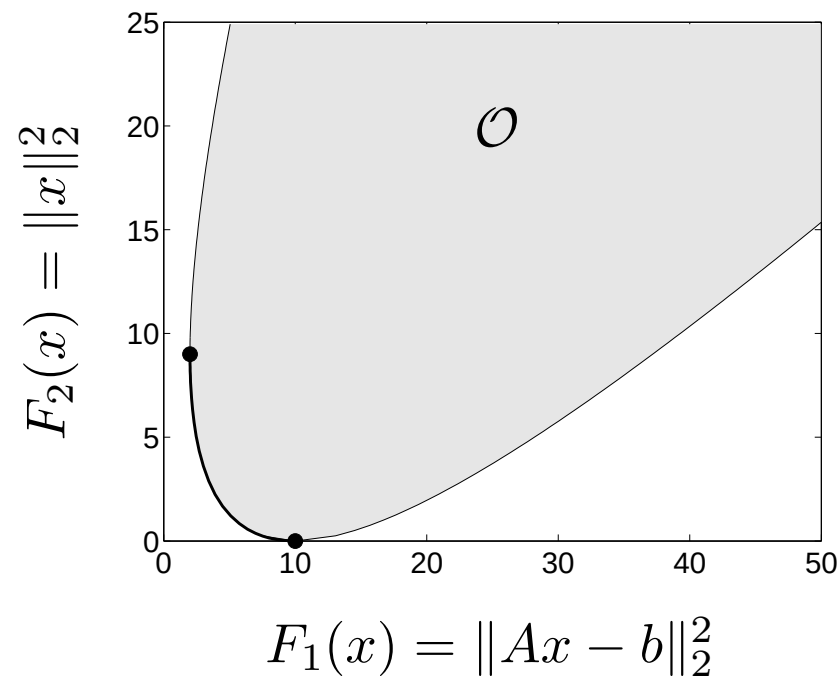
- feasible x^{po} is Pareto optimal if

$$y \text{ feasible, } f_0(y) \preceq f_0(x^{\text{po}}) \implies f_0(x^{\text{po}}) = f_0(y)$$

if there are multiple Pareto optimal values, there is a trade-off between the objectives

Regularized least-squares

$$\text{minimize (w.r.t. } \mathbf{R}_+^2) \quad (\|Ax - b\|_2^2, \|x\|_2^2)$$



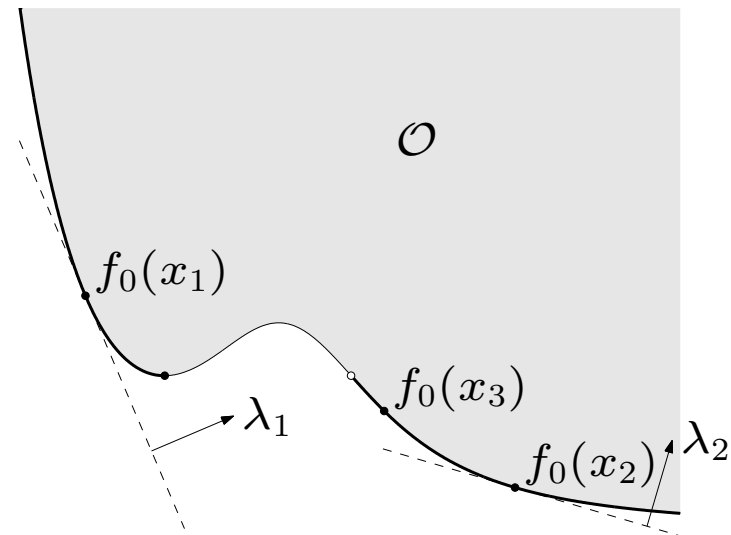
example for $A \in \mathbf{R}^{100 \times 10}$; heavy line is formed by Pareto optimal points

Scalarization

to find Pareto optimal points: choose $\lambda \succ_{K^*} 0$ and solve scalar problem

$$\begin{array}{ll} \text{minimize} & \lambda^T f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p \end{array}$$

if x is optimal for scalar problem,
then it is Pareto-optimal for vector
optimization problem



for convex vector optimization problems, can find (almost) all Pareto
optimal points by varying $\lambda \succ_{K^*} 0$

Scalarization for multicriterion problems

to find Pareto optimal points, minimize positive weighted sum

$$\lambda^T f_0(x) = \lambda_1 F_1(x) + \cdots + \lambda_q F_q(x)$$

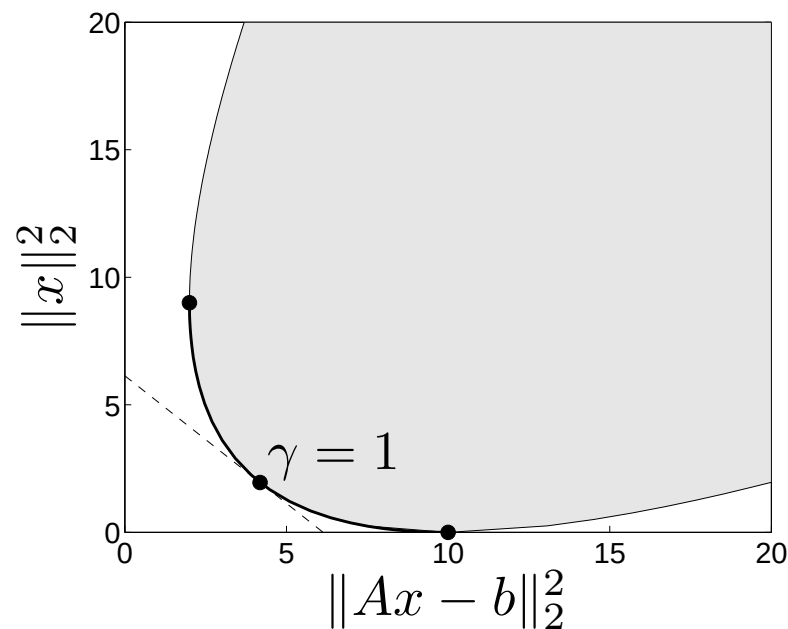
examples

- regularized least-squares problem of page 4–43

take $\lambda = (1, \gamma)$ with $\gamma > 0$

minimize $\|Ax - b\|_2^2 + \gamma \|x\|_2^2$

for fixed γ , a LS problem



Scalarized problem

$$\text{minimize} \quad \|Ax - b\| + \gamma \|x\|$$

- solution for $\gamma > 0$ traces out optimal trade-off curve
- other common method: minimize $\|Ax - b\|^2 + \delta \|x\|^2$ with $\delta > 0$

Tikhonov regularization

$$\text{minimize} \quad \|Ax - b\|_2^2 + \delta \|x\|_2^2$$

can be solved as a least-squares problem

$$\text{minimize} \quad \left\| \begin{bmatrix} A \\ \sqrt{\delta} I \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2$$

$$\text{solution } x^* = (A^T A + \delta I)^{-1} A^T b$$

Smoothing regularization

Adding other regularization terms

- Instead of $\|x\|$, we can use $\|Dx\|$;
- If D is diagonal, simply different weight on the components of x ;
- D can be used to capture smoothness.

Optimal input design

linear dynamical system with impulse response h :

$$y(t) = \sum_{\tau=0}^t h(\tau)u(t-\tau), \quad t = 0, 1, \dots, N$$

input design problem: multicriterion problem with 3 objectives

1. tracking error with desired output y_{des} : $J_{\text{track}} = \sum_{t=0}^N (y(t) - y_{\text{des}}(t))^2$
2. input magnitude: $J_{\text{mag}} = \sum_{t=0}^N u(t)^2$
3. input variation: $J_{\text{der}} = \sum_{t=0}^{N-1} (u(t+1) - u(t))^2$

track desired output using a small and slowly varying input signal

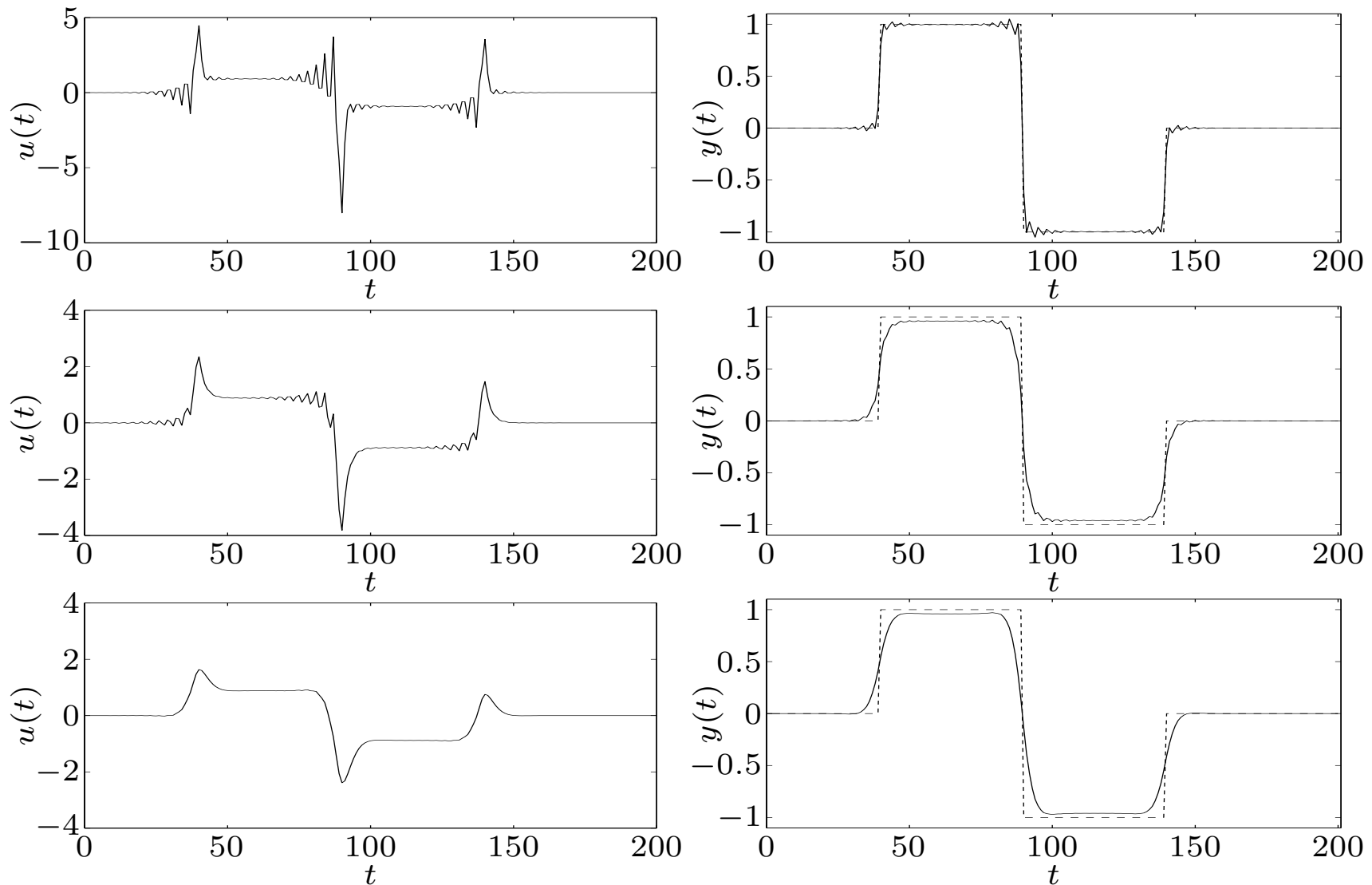
regularized least-squares formulation

$$\text{minimize} \quad J_{\text{track}} + \delta J_{\text{der}} + \eta J_{\text{mag}}$$

for fixed δ, η , a least-squares problem in $u(0), \dots, u(N)$

example: 3 solutions on optimal trade-off surface

(top) $\delta = 0$, small η ; (middle) $\delta = 0$, larger η ; (bottom) large δ



Signal reconstruction

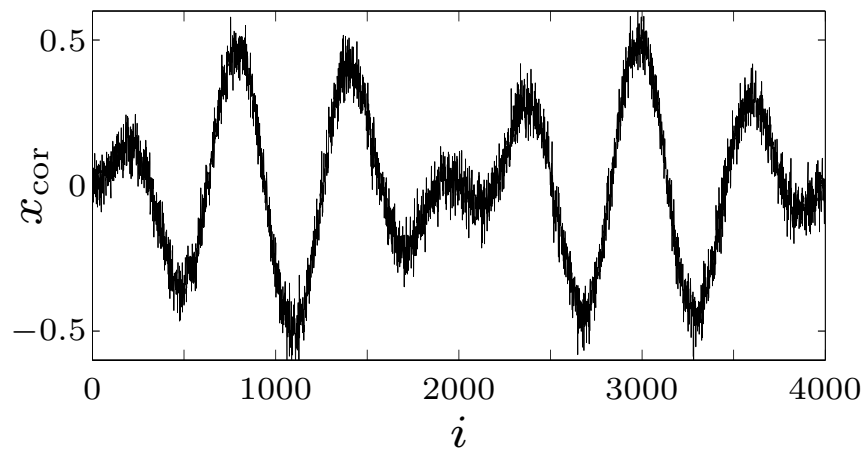
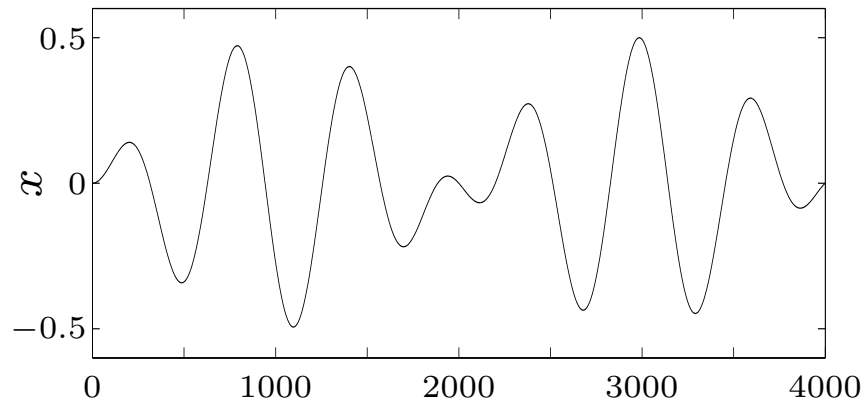
$$\text{minimize (w.r.t. } \mathbf{R}_+^2) \quad (\|\hat{x} - x_{\text{cor}}\|_2, \phi(\hat{x}))$$

- $x \in \mathbf{R}^n$ is unknown signal
- $x_{\text{cor}} = x + v$ is (known) corrupted version of x , with additive noise v
- variable $\hat{x}$ (reconstructed signal) is estimate of x
- $\phi : \mathbf{R}^n \rightarrow \mathbf{R}$ is regularization function or smoothing objective

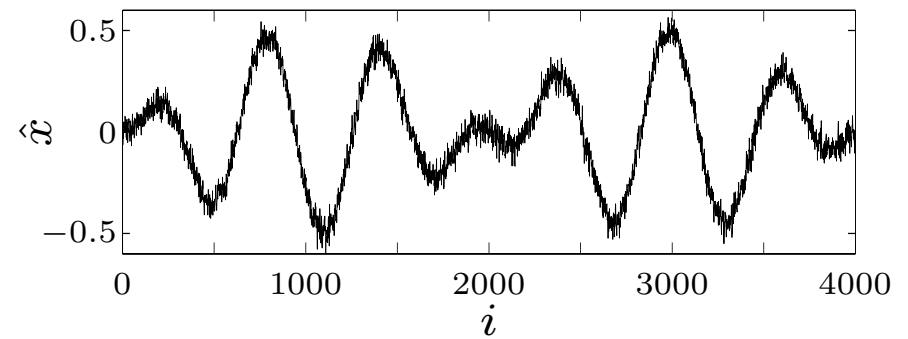
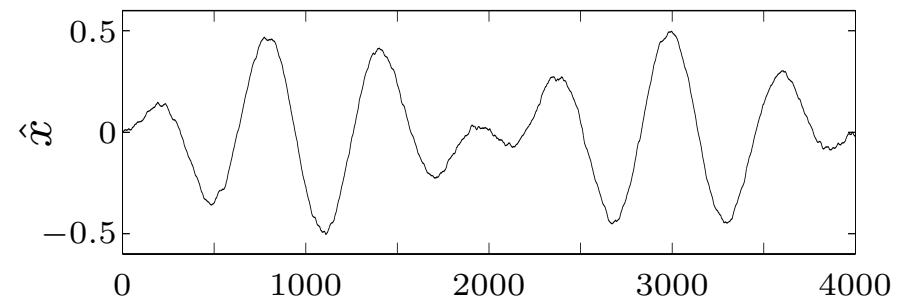
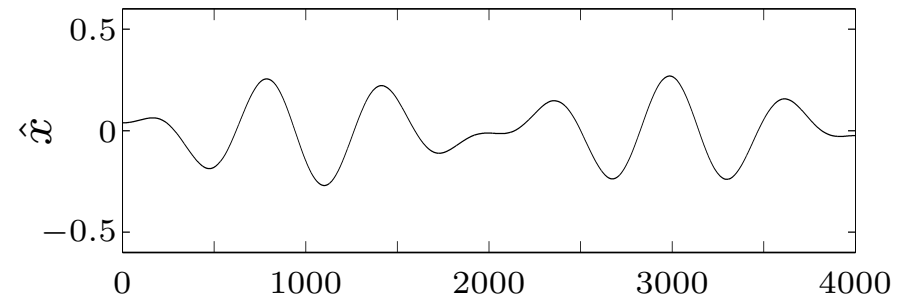
examples: quadratic smoothing, total variation smoothing:

$$\phi_{\text{quad}}(\hat{x}) = \sum_{i=1}^{n-1} (\hat{x}_{i+1} - \hat{x}_i)^2, \quad \phi_{\text{tv}}(\hat{x}) = \sum_{i=1}^{n-1} |\hat{x}_{i+1} - \hat{x}_i|$$

quadratic smoothing example

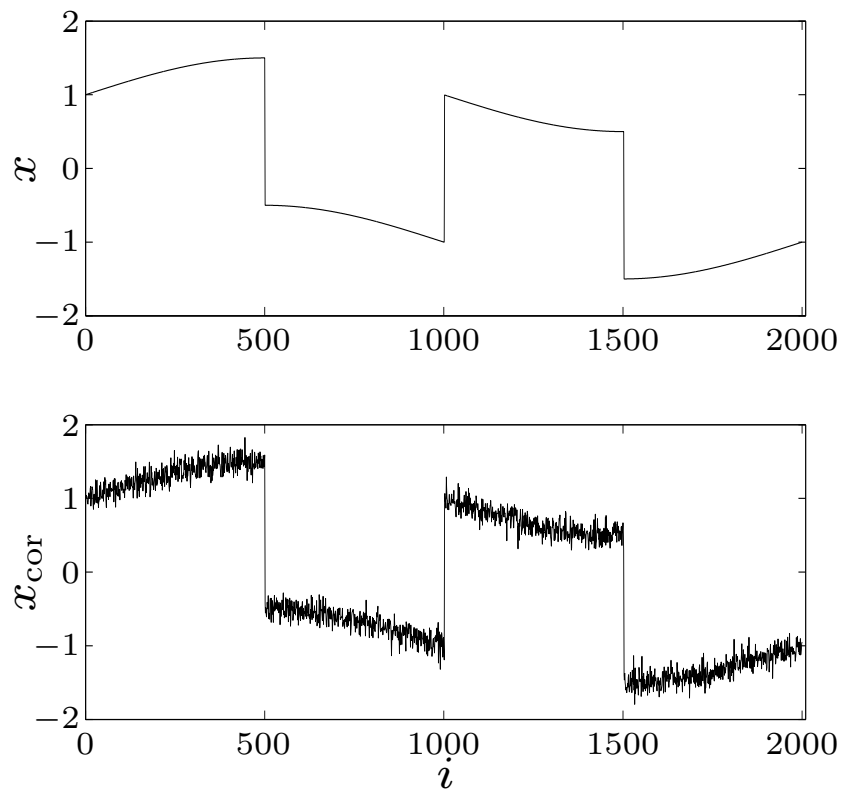


original signal x and noisy
signal x_{cor}

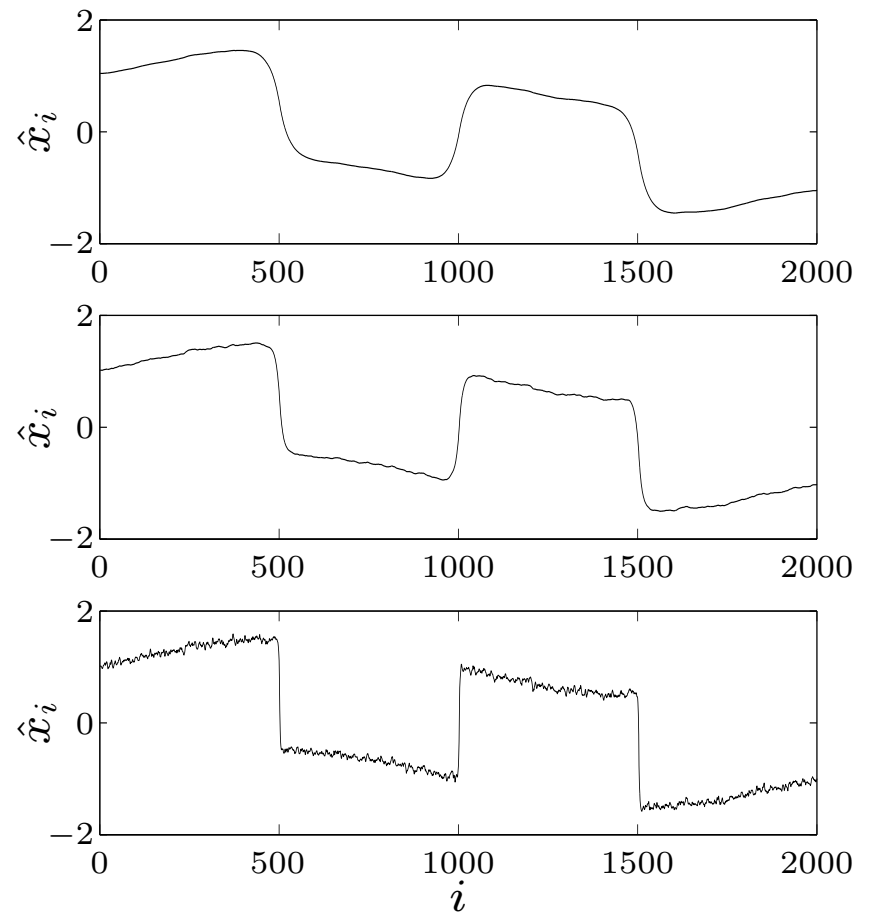


three solutions on trade-off curve
 $\|\hat{x} - x_{\text{cor}}\|_2$ versus $\phi_{\text{quad}}(\hat{x})$

total variation reconstruction example

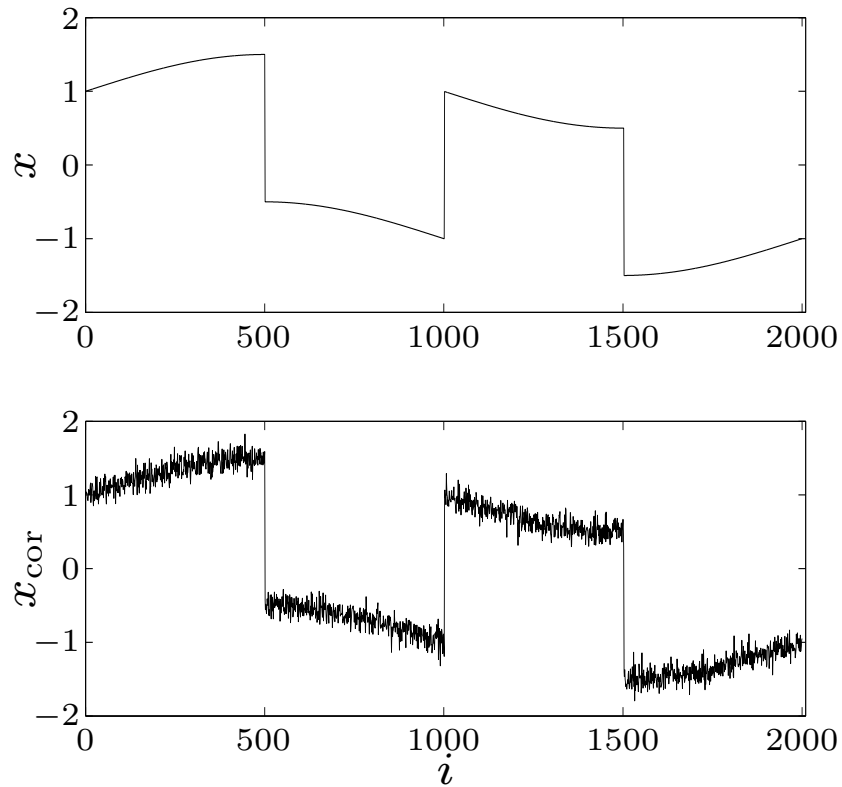


original signal x and noisy
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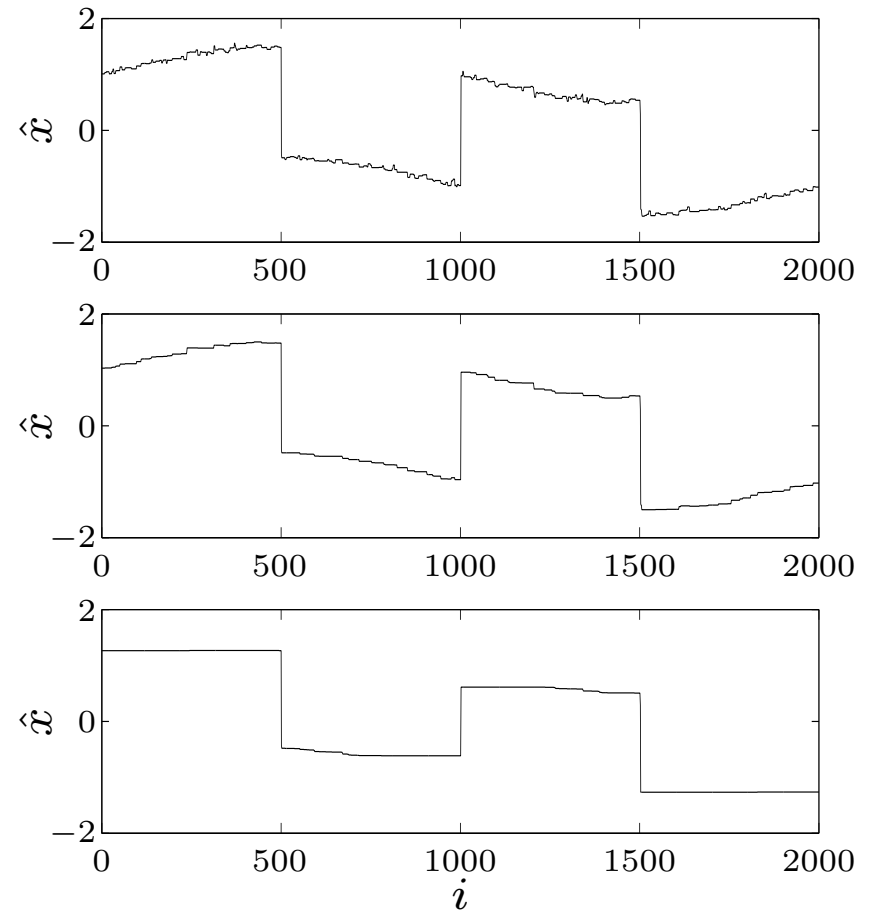


three solutions on trade-off curve
 $\|\hat{x} - x_{\text{cor}}\|_2$ versus $\phi_{\text{quad}}(\hat{x})$

quadratic smoothing smooths out noise **and** sharp transitions in signal



original signal x and noisy
signal x_{cor}



three solutions on trade-off curve
 $\|\hat{x} - x_{\text{cor}}\|_2$ versus $\phi_{\text{tv}}(\hat{x})$

total variation smoothing preserves sharp transitions in signal

ℓ_1 -norm regularization

Minimizing ℓ_1 -norm produces sparse solution (heuristically)

The problem

$$\begin{aligned} &\text{minimize} \quad \|Ax - b\| \\ &\text{subject to:} \quad \|x\|_0 \leq p. \end{aligned}$$

and the problem

$$\begin{aligned} &\text{minimize} \quad \|x\|_0 \\ &\text{subject to:} \quad \|Ax - b\| \leq d. \end{aligned}$$

Maybe approximately solved by solving

$$\text{minimize:} \quad \|Ax - b\|_2 + \gamma \|x\|_1,$$

Robust approximation

minimize $\|Ax - b\|$ with uncertain A

two approaches:

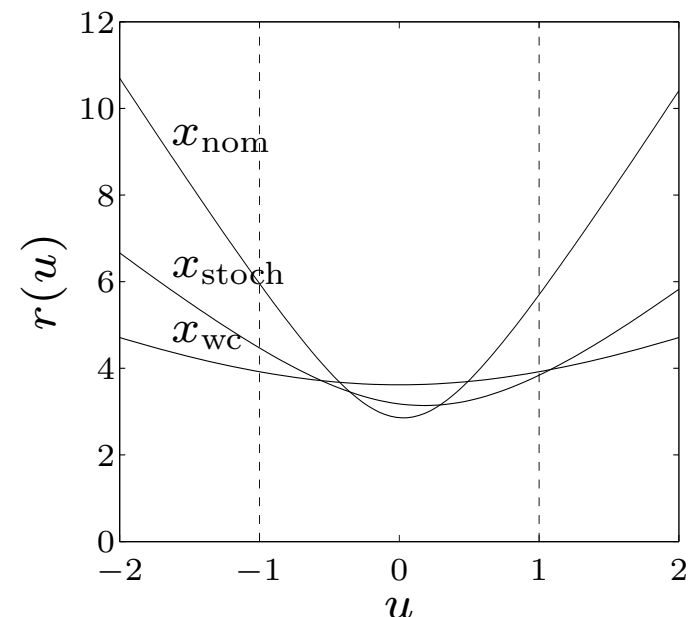
- **stochastic:** assume A is random, minimize $\mathbf{E} \|Ax - b\|$
- **worst-case:** set $\mathcal{A}$ of possible values of A , minimize $\sup_{A \in \mathcal{A}} \|Ax - b\|$

tractable only in special cases (certain norms $\|\cdot\|$, distributions, sets $\mathcal{A}$)

example: $A(u) = A_0 + uA_1$

- x_{nom} minimizes $\|A_0x - b\|_2^2$
- x_{stoch} minimizes $\mathbf{E} \|A(u)x - b\|_2^2$
with u uniform on $[-1, 1]$
- x_{wc} minimizes $\sup_{-1 \leq u \leq 1} \|A(u)x - b\|_2^2$

figure shows $r(u) = \|A(u)x - b\|_2$



Robust approximation: A simple case

The probability distribution has finite support

- $\text{prob}(A = A_i) = p_i, i = 1, 2, \dots, k;$

$$\text{minimize: } p_1 \|A_1 x - b\| + p_2 \|A_2 x - b\| + \dots + p_k \|A_k x - b\|$$

$\Downarrow$

$$\text{minimize: } p^T t$$

$$\text{subject to: } \|A_i x - b\| \leq t_i, i = 1, 2, \dots, k.$$

- If the norm is ℓ_2 , then we have an SOCP;
- If the norm is ℓ_1 or ℓ_∞ , then we have an LP.